



Demo e-commerce app testing

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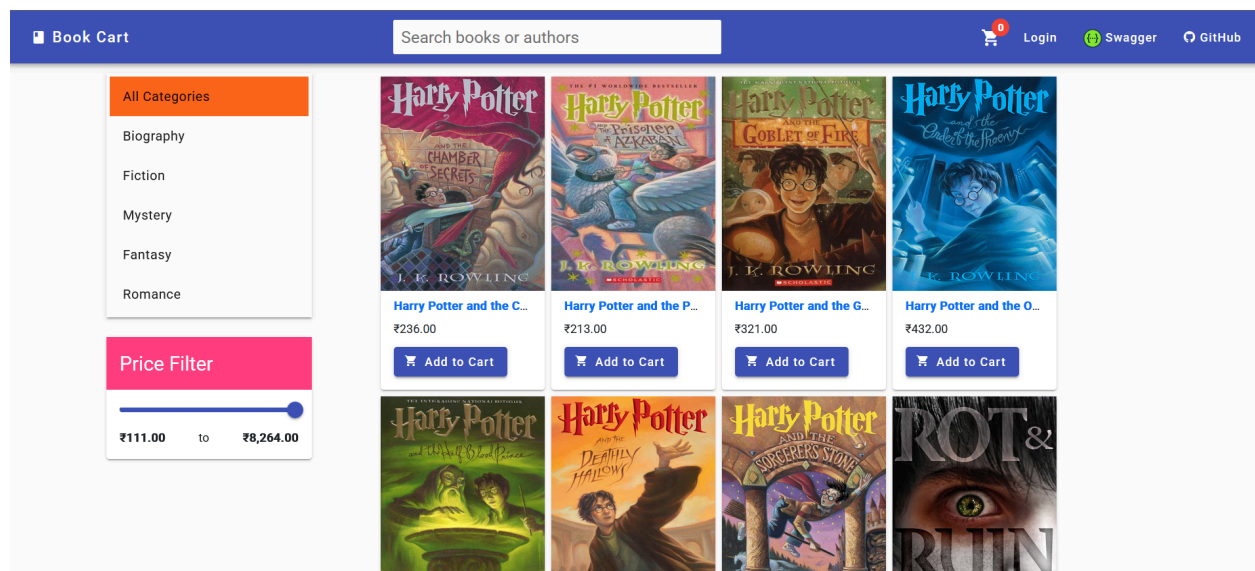
See automated smoke test and bug reports on this github repository:

https://github.com/aljaljak2/Bookcart-Test_project

About app:

Bookcart is an application for selling books online. It includes some important features like:

- 1) Register/Log in
- 2) Searching products and filtering by price and category
- 3) Adding to cart and editing cart
- 4) Reviewing book details and generating plot summary using Gemini AI
- 5) Checking and placing orders
- 6) Adding books to Wishlist

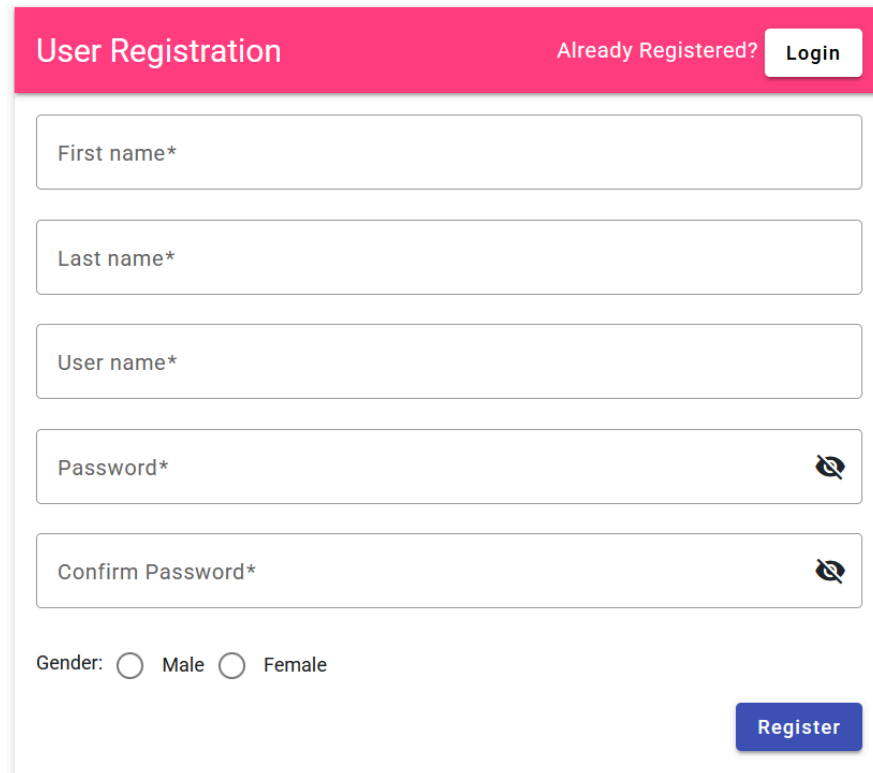


1. Features testing:

1.1 Register/Log in

1.1.1 Register

Here is how register form looks like:



The image shows a 'User Registration' form. At the top, there is a pink header bar with the text 'User Registration' on the left and 'Already Registered? Login' on the right. Below the header, there are five input fields: 'First name*', 'Last name*', 'User name*', 'Password*', and 'Confirm Password*'. The 'Password*' and 'Confirm Password*' fields have an eye icon to the right, indicating a toggle for password visibility. Below the input fields, there is a 'Gender:' label followed by two radio buttons, 'Male' and 'Female'. At the bottom right of the form, there is a blue 'Register' button.

Equivalence partitioning method will be used for the testing this form. Valid and invalid partitions will be identified in table on the next page.

Field	Valid partitions	Invalid partitions
First name	Only letters (John)	- Starts with number (1John) - Contains special character (John!)

		• Empty ("")
Last name	• Only letters (Doe)	• Starts with number (1Doe) • Contains special character (Doe!) • Empty ("")
User name	• Alphanumeric, length 3–20 (jack123)	• Too short (<3 chars) (aE) • Contains special characters (<500) • Empty ("")
Password	• have minimum 8 characters, at least 1 uppercase letter, 1 lowercase letter and 1 number (Password1)	• Less than 8 characters (Pass1) • No uppercase letters (password1) • No lowercase letters (PASSWORD1) • No digits (Password) • Empty password field ("")
Confirm password	• Matches password (Password1)	• Doesn't match password (Password123)
Gender	• Male or Female selected (male, female)	• Not selected

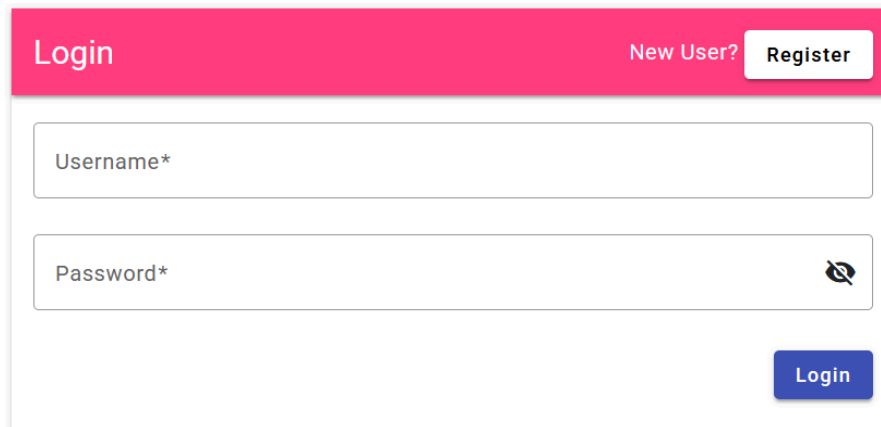
Next test cases are identified based on this valid and invalid partitions.

TCID	First name	Last name	Username	Password	Confirm password	Gender	Expected result	Notes
TC1	John	Doe	jack123	Password1	Password1	Male	Sucess	All fields valid
TC2	1John	Doe	jack123	Password1	Password1	Male	Error	Invalid first name (starts with digit)
TC3	John	Doe!	jack123	Password1	Password1	Male	Error	Invalid last name (special char)
TC4	John	Doe	aE	Password1	Password1	Male	Error	Username too short
TC5	John	Doe	jack\$123	Password1	Password1	Male	Error	Username contains special char
TC6	John	Doe	jack123	Pass1	Pass1	Male	Error	Password too short
TC7	John	Doe	jack123	password1	password1	Female	Error	Password missing uppercase
TC8	John	Doe	jack123	PASSWORD1	PASSWORD1	Female	Error	Password missing lowercase
TC9	John	Doe	jack123	Password	Password	Female	Error	Password missing digit

TC10	John	Doe	jack123	Password1	Password123	Female	Error	Password mismatch
TC11	John	Doe	jack123	Password1	Password1	-	Error	Gender not selected
TC12	""	""	""	""	""	-	Error	All fields empty

1.1.2 Log in

Now log in is going to be tested:

A login form with a pink header bar. The header bar contains the word "Login" on the left and "New User? Register" on the right. Below the header, there are two input fields: "Username*" and "Password*". The "Password*" field has a toggle icon (an eye with a slash) to its right. At the bottom right of the form is a blue "Login" button.

Login

New User? Register

Username*

Password* 

Login

Let's divide test cases for this feature in three groups:

- **Positive Test Cases**
- **Negative Test Cases – Invalid Combinations**
- **Edge/Usability Test Cases**

Positive Test cases:

ID	Test Case Description	Input Username	Input Password	Expected Result
TC1	Login with valid credentials	jack123	Password1	Login successful, redirect to home/dashboard

Negative Test Cases – Invalid Combinations:

ID	Test Case Description	Input Username	Input Password	Expected Result
TC2	Invalid username, valid password	jack123	Password1	Show "Invalid credentials"
TC3	Valid username, wrong password	jack123	Password2	Show "Invalid credentials"
TC4	Wrong username and wrong password	pertla	pertla123	Show "Invalid credentials"
TC5	Empty username	""	Password1	Show validation error (e.g. "Username required")
TC6	Empty password	jack123	""	Show validation error (e.g. "Password required")
TC7	Both fields empty	""	""	Show validation errors for both fields
TC8	SQL Injection attempt	jack123' OR '1'='1	Password1	Show "Invalid credentials" or handle gracefully
TC9	HTML/JS injection in username	<script>alert(1)</script>	Password1	Show "Invalid credentials" or sanitize input
TC10	HTML/JS injection in password	jack123	<script>alert(1)</script>	Show "Invalid credentials" or sanitize input
TC11	Username with leading/trailing spaces	" jack123"	Password1	Show "Invalid credentials" (or trim input if supported)
TC12	Password with leading/trailing spaces	jack123	" Password1"	Show "Invalid credentials" (or trim input if supported)

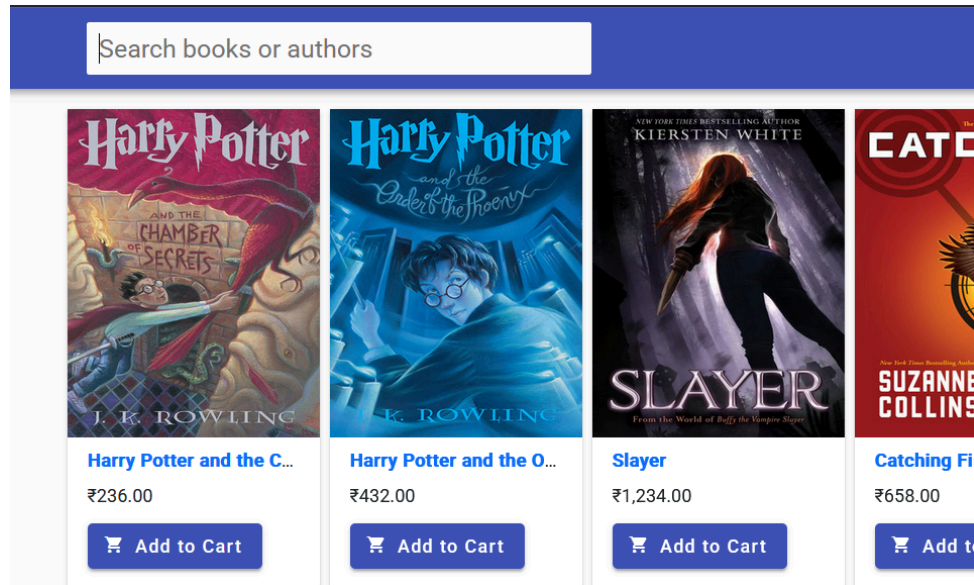
Edge/Usability Test Cases:

ID	Test Case Description	Input Username	Input Password	Expected Result
TC13	Max character length username (e.g. 50+ chars)	a...a	Password1	Show "Invalid credentials" or validation error
TC14	Max character length password (e.g. 50+ chars)	jack123	a...a	Show "Invalid credentials" or validation error
TC15	Login with username in different case	Jack123	Password1	Show "Invalid credentials" (or accept if case-insensitive)
TC16	Toggle password visibility icon	jack123	Password1	Password visibility toggles when icon is clicked
TC17	Press Enter key instead of clicking Login	jack123	Password1	Login is triggered successfully
TC18	Tab key focuses fields in correct order	-	-	Cursor should move: Username → Password → Login button

1.2 Searching products and filtering by price and category

1.2.1 Search bar

Search bar will be tested first.



Positive test cases:

ID	Test Case description	Steps to Reproduce	Expected Result
TC1	Search for full book title	Type "Harry Potter and the Chamber of Secrets" in the search bar and press Enter	The exact matching book appears in the results
TC2	Search for partial book title	Type "Harry Potter" and press Enter	All books with "Harry Potter" in the title appear
TC3	Search by author name	Type "J.K. Rowling" and press Enter	All books by J.K. Rowling appear
TC4	Search is case-insensitive	Type "sLayEr" and press Enter	Book "Slayer" appears despite case differences
TC5	Search using	Type "Chamber" and press Enter	"Harry Potter and

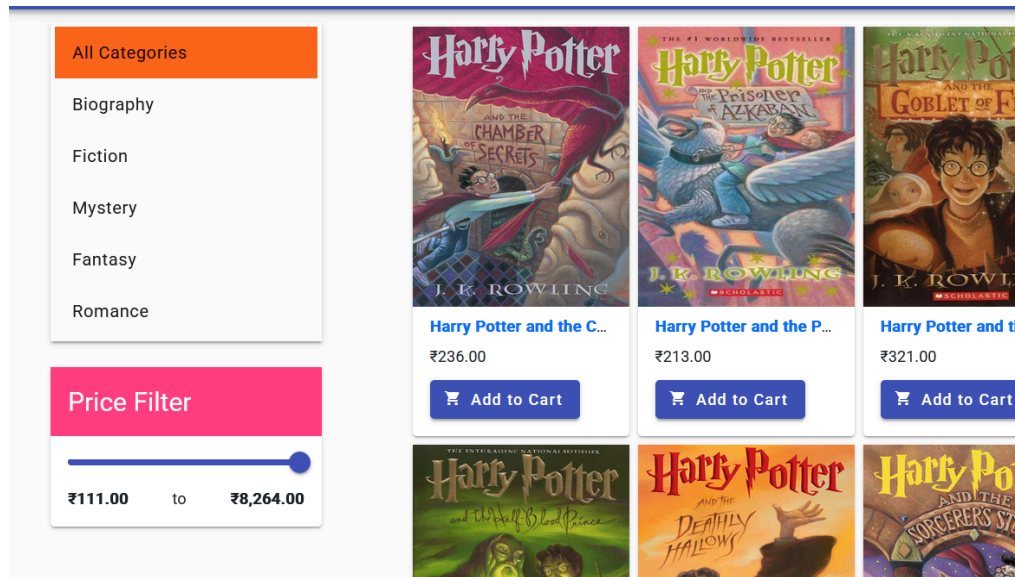
	keyword from subtitle		the Chamber of Secrets" is shown
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Negative test cases:

ID	Test Case description	Steps to Reproduce	Expected Result
TC6	Search with invalid string	Type "asdfghjkl" and press Enter	No results found message is shown
TC7	Search with only spaces	Type 5+ space characters and press Enter	Input ignored or no results found
TC8	Search with special characters	Type "@#\$%^&*()" and press Enter	Input handled gracefully; no crash or result
TC9	Leave search bar empty and press Enter	Click search with empty input	Page does not refresh or shows default book list
TC10	SQL Injection attempt	Type ' OR 1=1 -- in the search bar and press Enter	App does not return all results or crash; input is sanitized
TC11	Script injection attempt	Type <code><script>alert('XSS')</script></code> in the search bar and press Enter	Input is either escaped or ignored; no alert box or code execution

1.2.2 Price and category filters

Now price and category filters are going to be tested.



Category Filter - Test Cases:

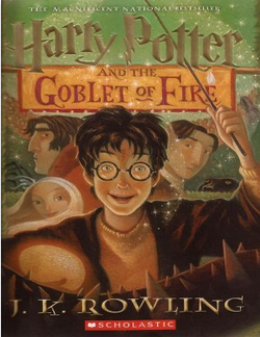
ID	Test Case description	Steps to Reproduce	Expected Result
TC1	Select a category "Biography"	Click on "Biography" in the category list	Only biography books are displayed
TC2	Select a category "Fiction"	Click on "Fiction" in the category list	Only fiction books are displayed
TC3	Select a category "Mystery"	Click on "Mystery" in the category list	Only mystery books are displayed
TC4	Select a category "Fantasy"	Click on "Fantasy" in the category list	Only fantasy books (like Harry Potter) are displayed
TC5	Select a category "Romance"	Click on "Romance" in the category list	Only romance books are displayed

TC6	Select “All Categories”	Click “All Categories”	All books are shown regardless of genre
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Price Filter - Test Cases

ID	Test Case description	Steps to Reproduce	Expected Result
TC7	Apply price filter with narrow range	Drag sliders to set range ₹200 to ₹300	Only books within ₹200–₹300 are shown
TC8	Set filter to full price range	Drag sliders to minimum and maximum ends	All books are shown
TC9	Set filter to a range with no matching books	Set range ₹7000–₹8000	No books displayed or “No results found” message shown

1.3 Adding to cart and editing cart



Harry Potter and the G...

₹321.00

 Add to Cart

Shopping cart Clear cart

Image	Title	Price	Quantity	Total	Action
	Harry Potter and the Goblet of Fire	₹321.00	- 1 +	₹321.00	
Cart Total:				₹321.00	CheckOut

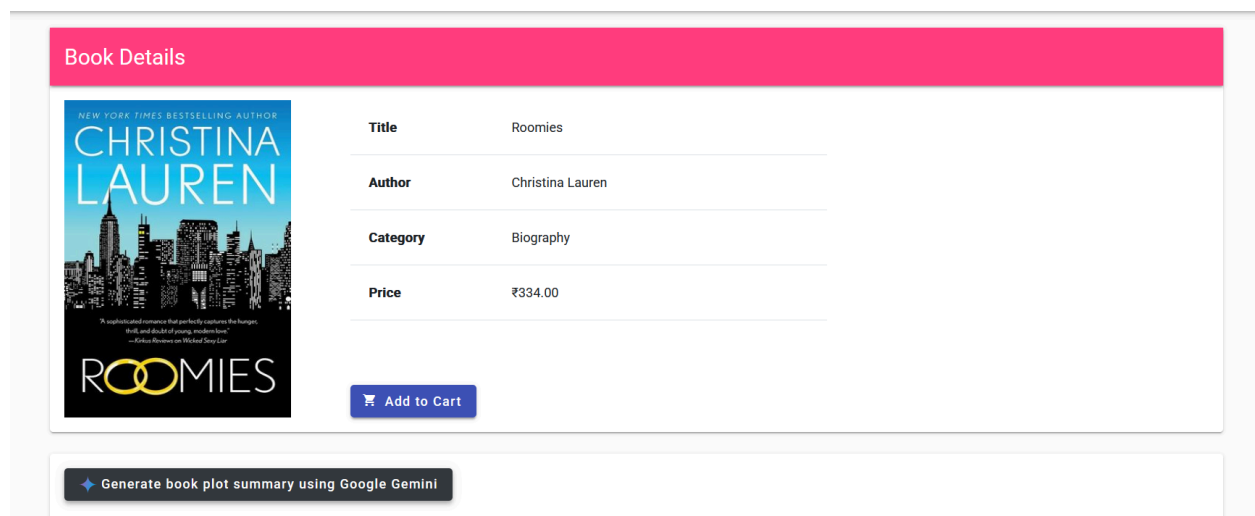
Add to Cart - Test Cases

ID	Test Case description	Steps to Reproduce	Expected Result
TC1	Add a single item to cart	Click “Add to Cart” on a book	Cart badge increments by 1; book appears in cart
TC2	Add the same item multiple times	Click “Add to Cart” on the same book multiple times	Quantity in cart reflects number of clicks; total price updates
TC3	Add multiple different books	Add two or more different books	Cart shows each book as a separate row with individual quantity and total

Edit Cart - Test Cases

ID	Test Case description	Steps to Reproduce	Expected Result
TC4	Increase item quantity in cart	Click the  button on an item	Quantity increases; total and cart total update
TC5	Decrease item quantity in cart	Click the  button	Quantity decreases; total updates; item removed if quantity hits zero
TC6	Remove item using trash icon	Click  icon beside an item	Item is removed from cart; cart total updates
TC7	Clear cart	Click “Clear cart” button	All items removed; cart becomes empty
TC8	Verify cart total calculation	Add 2+ items, adjust quantities	Cart total = $\Sigma(\text{item price} \times \text{quantity})$ — must be accurate

1.4 Reviewing book details and generating plot summary using Gemini AI



Test cases:

ID	Test Case description	Steps to Reproduce	Expected Result
TC1	View book title, author, category, price	Navigate to book details page	All book details are displayed correctly
TC2	Book image is displayed	Check for presence of book image	Image appears properly next to book metadata

TC3	Add book to cart from details page	Click "Add to Cart" button	Cart icon updates with item count; book is added to cart
TC4	Add same book again	Click "Add to Cart" multiple times	Quantity in cart increases with each click
TC5	Generate plot summary	Click "Generate book plot summary using Google Gemini"	AI-generated summary is displayed below the button

1.5 Checking and placing orders

1.5.1 General test cases

1) User is already logged in

Test cases:

ID	Test Case description	Steps to Reproduce	Expected Result
TC1	Canceling order	- Clicking on some book - Clicking "Add to cart" button - Clicking on cart icon on the navbar - Clicking on "CheckOut" button - Clicking Cancel on CheckOut page	It needs to return on Shopping cart page
TC2	Placing order	- Clicking on some book - Clicking "Add to cart" button - Clicking on cart icon on the navbar	Cart is cleared and order is placed successfully

		• Clicking on “CheckOut” button • Filling Shipping Address data • Clicking on “Place Order”	
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1.5.2 Test cases about filling this Check Out form

Check Out

Shipping address

Name*

Address Line 1*

Address Line 2*

Pincode*

State*

Place Order Cancel

Order Summary

Title	Quantity	Price	Total
Harry Potter and the Goblet of Fire	1	₹321.00	₹321.00
Grand Total			₹321.00

For this purpose Equivalence partitioning is used:

Field	Valid Partition	Invalid Partition
Name	Non-empty, alphabetic or alphanumeric string	Empty, only special characters, overly long string
Address Line 1	Non-empty, reasonable length	Empty, overly long

Address Line 2	Non-empty, reasonable length	Empty, overly long
Pincode	6-digit number, does not start with 0	Less than 6 digits, more than 6 digits, starts with 0, non-numeric
State	Non-empty, valid string	Empty, invalid characters

Test cases set assuming this equivalence partitions are:

TCID	Description	Input Values	Expected Result
TC1	All valid fields	Name: "John Doe", Addr1: "Street 123", Addr2: "Apt 4B", Pincode: "123456", State: "Maharashtra"	Order placed successfully
TC2	Empty Name	<i>(Name empty)</i> , valid others	Error on Name: "Name is required"
TC3	Empty Address Line 1	<i>(Addr1 empty)</i> , valid others	Error on Address Line 1
TC4	Invalid Pincode (starts with 0)	Pincode: "012345", valid others	Error on Pincode
TC5	Invalid Pincode (non-numeric)	Pincode: "123A56", valid others	Error on Pincode
TC6	Invalid Pincode (too short)	Pincode: "1234", valid others	Error on Pincode
TC7	Empty State	<i>(State empty)</i> , valid others	Error on State
TC8	All fields empty	(All blank)	All fields show required field errors

1.6 Adding books to Wishlist

User need to be already logged in.

Test cases:

ID	Test Case description	Steps to Reproduce	Expected Result
TC1	Add a book to the wishlist	• Log in as an existing user • Navigate to the book details page • Click the "Add to Wishlist" button	The book is added to the user's wishlist and is visible in the wishlist page
TC2	Remove a book from the wishlist	• Ensure the wishlist contains at least one book • Navigate to the wishlist page • Click the "Remove from Wishlist" button for any book	The selected book is removed from the wishlist
TC3	Add a wishlist item to the cart	• Ensure a book is present in the wishlist • Navigate to the wishlist page • Click "Add to Cart" button for that book	The book is added to the shopping cart
TC4	Clear all books from the wishlist	• Add multiple books to the wishlist • Navigate to the wishlist page • Click the "Clear	All books are removed from the wishlist

		Wishlist" button	
TC5	Wishlist retains data after navigation	• Add one or more books to the wishlist • Navigate to a different page (e.g., home or book list) • Return to the wishlist page	The books previously added to the wishlist are still present

2. Smoke test

Smoke test for this app will include Log in Positive Test case and Checking and placing orders Test case 1 in combination with Positive Test case for Filling CheckOut form. So, Smoke test flow looks like this:

ID	SMK_TC01
Title	End-to-End Smoke Test: Login → Add to Cart → Checkout → Place Order
Description	Verifies that a user can log in, add a book to the cart, proceed to checkout, and successfully place an order using valid form data. Also verifies that canceling from the checkout returns the user to the cart.
Precondition	User account exists with credentials: john123 / Password1

Steps to Reproduce:

1) Navigate to Login Page

2) Log In

- Username: **jack123**

- Password: **Password1**
- Click the **Login** button

3) Verify Login Success

- Confirm redirection to the home/dashboard page

4) Add Book to Cart

- Click on any book shown on the homepage or search result
- On the book details page, click **"Add to Cart"**

5) Start Checkout Process

- Click on the **Cart icon** in the navbar
- Click the **"CheckOut"** button

6) Cancel Checkout Flow

- On the checkout page, click **"Cancel"**
- Verify redirection back to the shopping cart page

7) Re-Start Checkout and Fill Valid Form

- Click **"CheckOut"** again
- Fill out the form:
 - Name: **John Doe**
 - Address Line 1: **Street 123**
 - Address Line 2: **Apt 4B**

- Pincode: 123456
- State: Maharashtra
- Click "**Place Order**"

8) Verify Order Placement

- A success message or confirmation screen appears

9) Log Out

- Click on the **Logout** button or link (usually in the navbar or dropdown)
- Verify redirection to the login or landing page

Expected Result

- User is successfully logged in
- Book is added to cart and visible
- Canceling checkout returns to cart
- Order is successfully placed with valid details
- User is logged out and returned to login or landing page

3. Positive and negative tests

Test suit is divided by app features here. In every section there are positive and negative tests for each feature.

Let make breakdown of positive and negative tests for each section here:

1) Register/Log in

Register

TCID	Positive test	Negative test
TC1	+	
TC2		+
TC3		+
TC4		+
TC5		+
TC6		+
TC7		+
TC8		+
TC9		+
TC10		+
TC11		+
TC12		+

Login

Test cases for login are already divided into 3 sections:

- Positive Test cases (represents positive test cases)
- Negative Test Cases – Invalid Combinations (represents negative test cases)
- Edge/Usability Test Cases (represents negative test cases)

2) Searching products and filtering by price and category

Search bar

In this part of section test cases are already divided on positive and negative test cases.

In other two subsections **Price Filter - Test Cases** and **Category Filter - Test Cases** all tests cases are positive.

3) Adding to cart and editing cart

In this section all test cases are positive.

4) Reviewing book details and generating plot summary using Gemini AI

In this section all test cases are positive.

5) Checking and placing orders

This section is containing of two positive test cases for placing order.

In subsection **Test cases about filling this Check Out form** only first test case is positive. All other test cases are negative.

6) Adding books to wishlist

This sections contains of only positive test cases.

4. Bug report

Detailed bug reports with steps for reproduction are provided on github repository: https://github.com/aljaljak2/Bookcart-Test_project in form of github issues.

Which Test Cases from this document are failing?

Register: TC2, TC3, TC4, TC5

Search bar: TC3

Reviewing book details and generating plot summary using Gemini AI: TC5